

TECHNICAL RELEASE

Segovia v0.4.1: a bounded-memory, near-real-time streaming preprocessor for Neuropixels-scale electrophysiology

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Abstract

Segovia is an open-source Rust library with Python bindings for bounded-memory, near-real-time preprocessing of high-density extracellular electrophysiology recordings. It applies a bandpass-filter, common-median-reference, and whitening chain to Neuropixels-scale (384–768 channels, 30 kHz) 16-bit integer streams one chunk at a time, releasing the Python Global Interpreter Lock for each Rust computation. Peak memory is bounded by chunk size, not by recording length, and scales analytically. On a full 55.8-minute International Brain Laboratory AP-band recording (385 channels, mtscomp-compressed) at a 300 ms chunk budget, Segovia holds 99.7% real-time deadline adherence at 0.21 GB peak resident memory. The package ships three production readers (SpikeGLX, Zarr, mtscomp) and a built-in streaming synthetic simulator.

Availability and Implementation: Source code (Rust and Python, AGPL-3.0-or-later):

<https://github.com/fcarvajalbrown/Segovia>. Install via `pip install segovia` (PyPI) or `cargo add segovia` (crates.io). Version v0.4.1.

Key words: electrophysiology; Neuropixels; streaming; real-time preprocessing; bounded memory; Rust; spike sorting

Statement of Need

Neuropixels probes [1] acquire 384 channels at 30 kHz (~22 MB/s per probe). Standard preprocessing pipelines (SpikeInterface [2], MountainSort) run offline on completed recordings and are optimized for batch throughput. Near-real-time applications, such as closed-loop stimulation and online brain-machine interface decoding, require that each incoming chunk of samples be preprocessed within its acquisition period (the real-time deadline), with peak memory independent of recording length. Batch tools driven one chunk at a time re-read filter-margin neighbourhoods on every call and do not bound memory analytically. Segovia was built to fill this gap: a composable streaming preprocessor whose memory ceiling is fixed by chunk size and whose per-chunk latency meets the acquisition deadline on real recordings.

Implementation

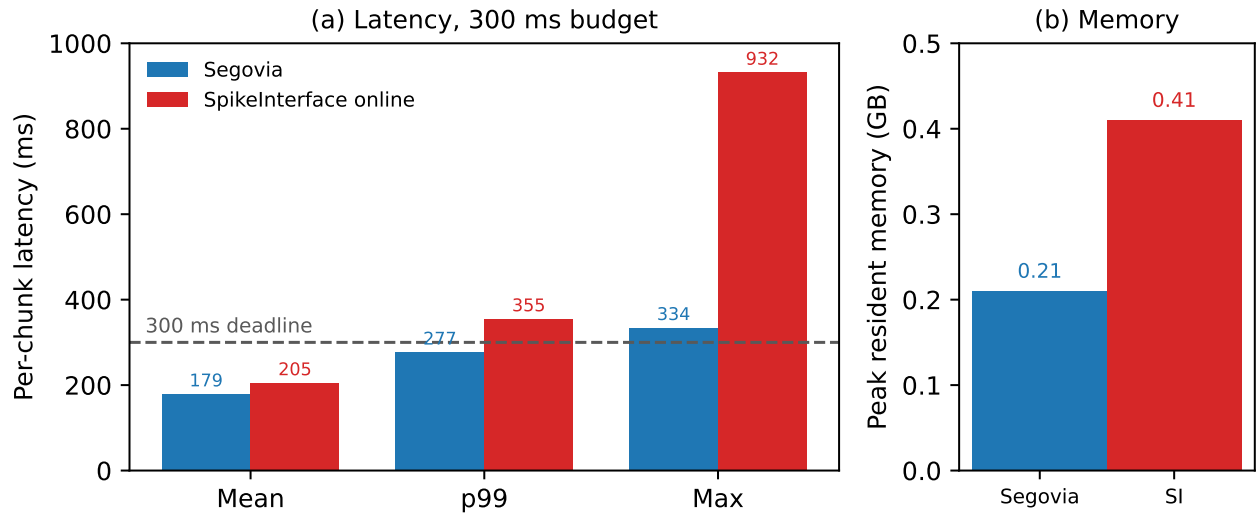


Figure 1. Full-length steady-state comparison on the real IBL AP-band recording (385 channels, 55.8 min, 300 ms budget). (a) Per-chunk latency (mean, p99, maximum) against the 300 ms real-time deadline; Segovia's maximum of 334.5 ms yields 99.7% deadline adherence, whereas SpikeInterface online reaches 932 ms. (b) Peak resident memory, 0.21 GB versus 0.41 GB.

Table 1. Real IBL AP-band recording, full length (385 channels, mtscomp-compressed, 55.8 min, 11,167 chunks, 300 ms budget, steady state).

Engine	Mean	p99	Max	Adherence	Peak RSS	Jitter
Segovia	179.2 ms	277.0 ms	334.5 ms	99.7%	0.21 GB	38.6 ms
SpikeInterface online	205.3 ms	355.0 ms	932.0 ms	94.7%	0.41 GB	60.5 ms

Core abstraction

Segovia's `ChunkSource` trait is an iterator over fixed-size (samples $\times$ channels) 16-bit integer buffers. Three production implementations are provided:

- `SpikeGlxReader`: memory-mapped SpikeGLX `.bin + .meta` (zero-copy).
- `ZarrReader`: chunked Zarr arrays (gzip, zstd, Blosc) via the `zarrs` crate.
- `CbinReader`: mtscomp-compressed IBL `.cbin`, per-chunk zlib decompression via `flate2`.

Preprocessing chain

`preprocess(chunk_source, config)` applies a Rayon-parallelized chain: a 5th-order Butterworth bandpass filter (zero-phase `sosfiltfilt`), common median reference, and global ZCA whitening. The Python GIL is released via `PyO3's allow_threads` for the Rust computation. Cross-chunk IIR filter state is maintained deterministically regardless of thread count.

Peak resident memory scales as $\text{batch} \times (\text{chunk} + 2 \text{ margin}) \times \text{channels} \times \text{sizeof}(f32)$ and is independent of recording length. The default (auto) batch width is capped at $\min(\text{logical threads}, 4)$ as a conservative out-of-memory guard, bounding the default footprint to ~ 1.2 GB on any machine; callers may pin an explicit batch. This cap is an out-of-memory safety guard, not a throughput-optimum claim.

Built-in simulator

`SyntheticEphysReader` emits arbitrarily long synthetic streams without writing to disk. Spike templates use a Ricker temporal waveform with point-source spatial decay ($V(r) = A d_{\perp}/r$); firing is Poisson per unit; noise is additive white Gaussian. The pseudorandom number generator (`SplitMix64 + xoshiro256++`) is platform-independent and chunk-size-independent. `ground_truth()` returns (sample, unit, channel) tuples for downstream sorter evaluation.

Results

Evaluation follows a replay-at-acquisition-rate methodology: data are streamed at the true 30 kHz rate with per-chunk compute latency measured by a monotonic clock. Deadline adherence is the fraction of chunks with latency at or below the chunk period.

Figure 1 and Table 1 report the full-length steady-state comparison. At steady state Segovia leads on every axis; the decisive margins are peak memory ($2\times$), maximum latency ($2.8\times$), and jitter. Table 2 is the cold-start first-60 s window, where SpikeInterface's warm-up cost is highest and the adherence gap is widest (100% vs 69.5% at 300 ms); that gap narrows to the steady-state figures over the full recording.

Table 2. Real IBL AP-band recording, cold-start first 60 s (385 channels, mtscomp-compressed).

Chunk	Engine	Mean latency	p99	Adherence	Peak RSS
100 ms	Segovia	92.9 ms	122.0 ms	74.2%	0.21 GB
100 ms	SpikeInterface online	112.0 ms	275.2 ms	64.2%	0.46 GB
300 ms	Segovia	194.5 ms	256.4 ms	100%	0.28 GB
300 ms	SpikeInterface online	245.8 ms	365.7 ms	69.5%	0.52 GB
1000 ms	Segovia	617.3 ms	705.9 ms	100%	0.49 GB
1000 ms	SpikeInterface online	786.0 ms	947.5 ms	98.2%	0.74 GB

On synthetic recordings (384 channels, `SyntheticEphysReader`, seed 0) Segovia reaches 100% deadline adherence at all chunk sizes (100/300/1000 ms) with jitter 3.6/8.6/18.6 ms.

Peak memory is bounded and file-size-independent on both sources; on the full 55.8-minute (29 GB compressed) real recording the memory bound holds to within 1% of a ten-minute slice. In a batch-throughput comparison on that full recording, Segovia at a pinned batch held peak memory below both SpikeInterface executor modes (1.19 GB vs 2.18 GB thread-pool and 4.42 GB process-pool) and completed in less wall time (806 s vs 923 s and 1022 s) in a single run; the memory bound is the robust result and the wall-time margin warrants multi-run replication. Throughput exceeds the 22 MB/s Neuropixels acquisition rate at all configurations.

Known limitation: the zero-phase Butterworth filter introduces a 50 ms look-ahead; a causal filter mode is not yet implemented. Benchmarks are on a single machine (Windows, 8 physical / 16 logical cores).

Availability of source code and requirements

- Project name: Segovia
- Project home page: <https://github.com/fcarvajalbrown/Segovia>
- Operating system(s): Linux, macOS, Windows (platform independent)
- Programming language: Rust (core, edition 2021, minimum supported Rust version 1.74); Python bindings via PyO3
- Other requirements: Python ≥ 3.8 for the bindings (prebuilt abi3 wheels on PyPI); a Rust toolchain and `maturin` to build from source
- License: AGPL-3.0-or-later (OSI-approved)
- RRID: to be registered at SciCrunch.org prior to publication
- bio.tools ID: not yet registered
- Version: v0.4.1

The source code is under the AGPL-3.0-or-later license (an Open Source Initiative approved license) and is openly available in the GitHub repository above, with versioned releases on PyPI and crates.io.

Availability of Test Data

Segovia is a software library; no new experimental dataset was generated. The real recording used for evaluation is a publicly released International Brain Laboratory AP-band Neuropixels recording (`_spikeglx_ephysData_g0_t0.imec0.ap.cbin`), obtainable from the International Brain Laboratory data portal. Synthetic recordings are generated deterministically on demand by `SyntheticEphysReader` (seed-fixed, no deposit required). The form of the required GigaDB deposit for a software Technical Release is being confirmed with the editorial team.

Declarations

List of abbreviations

AGPL: GNU Affero General Public License; AP: action potential (band); CMR: common median reference; GIL: Global Interpreter Lock; IBL: International Brain Laboratory; IIR: infinite impulse response; RSS: resident set size; SOS: second-order sections; ZCA: zero-phase component analysis (whitening).

Ethical Approval

Not applicable. No human or animal data were collected for this work; the recording used for evaluation is a publicly released dataset.

Competing Interests

The author declares that they have no competing interests.

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Author's Contributions

F.C.B. is the sole author and performed all work (CRediT: conceptualization, software, methodology, investigation, writing).

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